# Gullapalli Madhava Asrith Murthy

ai23btech11007@iith.ac.in | madhavaasrith.g@gmail.com | +91 9136000474 | LinkedIn

## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.47 |
| Fiitjee Junior College | Class XII (TSBIE) | 2023 | 96.20% |
| Ryan International School | Class X (ICSE ) | 2021 | 97.80% |

## WORK EXPERIENCE

**Intern | CGI**

- **Engineered** secure deletion workflows for an internal AI-powered hiring platform using **Python, SQLAlchemy, PostgreSQL, and Angular**.
- **Designed** backup and **logging schemas** to preserve deleted entities and their dependent records, ensuring audit compliance and enabling **data restoration** while maintaining database integrity.
- **Resolved** complex **foreign-key** and cascading-relationship constraints across user profiles, and assessment tools and reports to prevent **data orphans** during deletion process.
- **Implemented RBAC** and Angular UI controls for deletion operations, preventing **unauthorized deletion** through direct URL/API access.

## PROJECTS

**Document Management Repository** *(Coursework / Group Project / Prof. Manish Singh)*

- **Developed** a full-stack, AI-powered document management system using **React, Node.js, Express, MongoDB, Redis, and AWS S3**, supporting public, private, and organization-based document spaces.
- **Engineered** a multi-criteria **document filtering pipeline** using dynamic MongoDB aggregation queries to support **simultaneous searches** across multiple parameters, efficiently delivering filtered results at scale.
- **Implemented** a multi-layer **Redis caching** strategy with targeted invalidation on file modifications, ensuring strict **data consistency** across dashboard statistics, S3 presigned URLs, and AI tag results.
- **Integrated** a Groq LLM (Llama 3.3) **auto-tagging pipeline** for **semantic keyword extraction** and classification, utilizing SHA-256 hash-based deduplication to prevent redundant API calls for identical files.

**Inventory Management System** *(Coursework / Group Project / Prof. Shirshendu Das)*

- **Developed** an inventory management system as part of a team of four, focusing on robust **database design**.
- **Designed** normalized relational schemas and implemented 10+ relations with 1,000+ tuples in **MySQL**.
- **Implemented** indexes, triggers, and stored procedures for efficient data handling and automated stock updates, ensuring transactional integrity and concurrency control.
- Enforced **security** with role-based access control.
- Integrated with a web-based **MVC frontend** to demonstrate full-stack functionality.

**Financial Decision Maker** *(Coursework / Group Project / Prof. Maunendra Desarkar)*

- **Engineered** a financial time-series preprocessing pipeline for an autonomous **BUY / SELL / HOLD** trading system, handling cleaning, normalization, missing values, and temporal alignment across six assets.
- **Developed** price-based features including **5-day/10-day moving averages, RSI, historical volatility, and market momentum** from historical price sequences.
- **Built** the numerical feature representation used in a multimodal fusion pipeline combining market features with **FinBERT-derived financial sentiment embeddings** and fundamental signals.
- **Contributed** to the decision engine that converts fused model outputs into daily trading actions, with the system designed to optimize **risk-adjusted returns measured by Sharpe ratio**.

**RISC-V Simulator** *(Coursework / Prof. Rajesh Kedia)*

- **Built** an assembler and simulator for the RISC-V instruction set.
- **Engineered** an interactive command-line debugger supporting breakpoints, memory inspection, and call-stack tracing.
- **Supports** cache simulation and cache usage statistics, including configurable parameters and tracks hit/miss statistics.

## SKILLS

**Languages:** C++, Java, Python, SQL

**AI/ML:** NLP, Computer Vision, LLMs

**Frameworks/Libraries:** PyTorch, FastAPI, OpenCV, Spring Boot

**Databases:** MySQL, PostgreSQL

## RELEVANT COURSES

**AI**: Foundations of Machine Learning, Deep Learning, Computer Vision, Natural Language Processing

**CS**: Computer Architecture, Discrete Mathematics, DBMS, Data Structures and Applications, Algorithms, Software Engineering

**Mathematics**: Probability and Random Variables, Calculus, Vector Calculus, Differential Equations, Applied Statistics

**Electrical Engineering**: Linear Systems and Signal Processing, Matrix Theory

## EXTRACURRICULAR

**Volunteering:** National Service Scheme(NSS), IITH - Student Volunteer

**Hobbies:** Art, Cricket, Table Tennis